

2 Related Work

Vision-Language-Action Models. Vision-language-action (VLA) models have demonstrated strong generalization across diverse tasks and environments by transferring semantic priors from pretrained vision-language models to robotic manipulation [2, 22, 3, 4, 5, 6, 23, 24, 25, 26]. Their progress is further supported by large-scale robot datasets [27, 28], enabling policies to acquire broad visuomotor skills from heterogeneous demonstrations. However, most VLA approaches remain *policy-centric*: they map observations directly to actions, with temporal structure and physical dynamics only implicitly learned from action-labeled data. Consequently, physical dynamics are not treated as first-class modeling targets, which may limit data efficiency and robustness in long-horizon manipulation [29].

World Action Models. World action models (WAMs) provide a dynamics-centric alternative to direct observation-to-action policies by modeling how the world evolves in conjunction with robot actions. Early approaches first predict future visual goals and then infer actions from the current observation and the predicted visual goals [30, 31, 32, 33]. Other methods move toward unified video-action modeling, in which future observations and actions are learned jointly [34, 35, 36, 37, 13, 38, 17, 39, 12, 7, 11, 40]. Recently, several methods have shown that video prediction can serve primarily as training-time supervision for dynamics modeling, enabling inference without costly video denoising [14, 15]. However, most efficient WAMs rely on bounded recent context, whereas full-history methods incur inference latency and GPU memory cost that grow rapidly with sequence length [7, 11]. This limitation motivates MemoryWAM’s efficient persistent memory, which substantially accelerates inference while reducing GPU memory overhead.

Memory Mechanisms for Sequential Modeling. Memory is central to sequence modeling, motivating mechanisms ranging from recurrent neural networks (RNNs) [41] and long short-term memory (LSTM) [42] to Transformers with full attention [43], linear attention [44, 45, 46], and test-time training [47]. These designs make trade-offs among memory capacity, update efficiency, and retrieval fidelity. Similar memory-efficiency trade-offs have recently emerged in long-horizon visual systems, including streaming 3D reconstruction [48, 49, 50], long video generation [51, 52, 53, 54], and memory-dependent robotic manipulation [55, 1, 56, 57, 58, 59, 60, 61]. Our work studies memory for WAMs and introduces a hybrid memory mechanism that retains long-range context while substantially reducing GPU memory overhead and inference latency.

3 Method

3.1 Overview

Prior approaches in vision-language-action models (VLAs) [4, 62] and world action models (WAMs) [8, 14] typically map recent observations to actions, relying on a bounded temporal window $o_{t-N:t}$ and task instruction l :

$$a_t = \pi_{\text{short}}(o_{t-N:t}, l). \quad (1)$$

While effective for short-horizon tasks, they struggle in non-Markovian environments where decisions depend on long-range history.

A straightforward approach to retain long-term history is to preserve the full KV cache of all past observations within an autoregressive Transformer [7, 11, 17]. While this design allows the policy to access the complete history $o_{1:t}$, it suffers from rapidly increasing GPU memory usage and inference latency as t grows, making it impractical for long-horizon tasks. To overcome these efficiency limitations, we draw inspiration from human cognition: (1) **short-term memory** supports ongoing action planning but has limited capacity [19]; (2) **long-term memory** tends to encode abstract **gist traces** rather than verbatim details [20]; and (3) **event-boundary memory** emphasizes the state at the onset of a task [21]. Motivated by these insights, we propose a *hybrid memory* mechanism that preserves high-fidelity short-term context, maintains a small set of gist tokens summarizing long-range history, and retains anchor frames at task onset. This human-like hybrid memory significantly reduces inference cost while enabling the policy to reason over long-horizon dependencies.